

NEUR 1440: Astrocyte Regulation Confines Representational Drift to Null-space

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Abstract

Representational drift refers to the continuous change in neural activity patterns despite stable behavior. It poses a critical challenge to the long-term robustness of the intracortical Brain-Computer Interfaces (iBCIs). Moving beyond the single-neuron level, while the "Null-space Hypothesis" suggests that the brain confines this drift to non-coding dimensions, the physiological mechanism remains unclear. This paper proposes that astrocytes act as the primary "stabilizers" in cortical circuits. Here, I hypothesize that astrocytes function as local hubs that integrate and balance the activity of neural ensembles. By modulating intrinsic excitability and calcium signaling, astrocytes may dynamically determine a subset of active neurons within a redundant population, which causes the representational drift while maintaining a constant aggregate output in the low-dimensional readout. I review evidence suggesting astrocytes act as "neuron excitability schedulers" via GLT1-mediated glutamate clearance and propose an experimental framework to validate that disrupting astrocytic feedback uncouples behavioral stability from neural drift. Finally, as indicated in this theory that an astrocytic ensemble gives a stable encoding readout, I introduce a Neuron-Astrocyte Hybrid BCI (NAH-BCI) architecture. A major implication of this theory is that the integrated readout from specific astrocyte ensembles provides a significantly more stable encoding signal than individual spiking activity, offering a novel pathway to overcome decoder degradation.

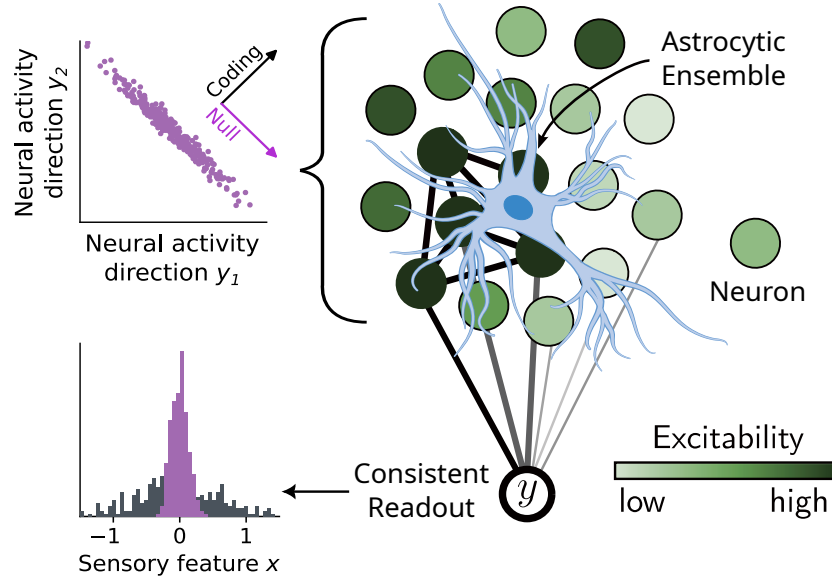


Figure 1: Summary plot: Astrocytes regulate neural ensembles' excitability for specific behavior encoding recruitment, causing the representational drift while keeping the drift in the null-space.

1 Representational Drift and Null-Hypothesis

1.1 Representational Drift

Representational drift refers to the phenomenon where the neural activity patterns associated with specific sensory stimuli, cognitive states, or motor behaviors change continuously over time, even when the external behavior remains stable. Traditionally, theories of neural coding assumed that the relationship between neural firing and specific variables, such as movement direction, was static. However, recent chronic recording studies have demonstrated that the neural representation is highly dynamic. Individual neurons may alter their tuning curves, drop out of the ensemble, or be replaced by other neurons in the population.

This phenomenon is widespread across cortical areas. In the primary visual cortex (V1), studies have demonstrated that while mice performed a consistent visual task, the cosine tuning of neurons responding to the stimuli of specific directions evolved over weeks, with only a small fraction of neurons maintaining stable tuning [5]. Similarly, in the primary motor cortex (M1), neurons typically exhibit directional cosine tuning, where the firing rate is modulated by the angle of movement related to a specific "preferred direction" [9]. Representational drift manifests here as the instability of these tuning parameters: a neuron's preferred direction may rotate, or its modulation depth may fluctuate significantly over days. Consequently, a neuron that encodes "leftward" movement on one day might shift to encode a different direction—or cease modulation entirely—in subsequent sessions [2]. A simple illustration of the representational drift in cosine tuning is shown below in Figure 2.

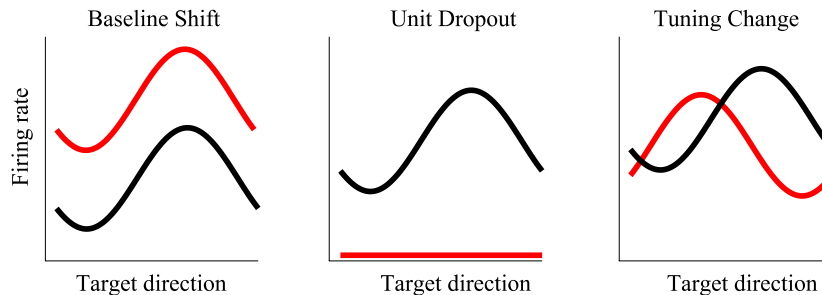


Figure 2: Example of representational drift in cosine tunings.

For intracortical Brain-Computer Interfaces (iBCIs), presentational drift is a primary source of signal non-stationarity, posing a critical hurdle to long-term system robustness. The phenomenon of representational drift presents three key challenges for the effective application of iBCIs:

1. **Decoder Degradation:** Most iBCI decoders, such as Kalman filters [20] or deep learning models [12], are trained on data collected during a specific session. Because the underlying neural distribution shifts over days or weeks due to representational drift, a static decoder inevitably loses accuracy [11]. The mapping learned on "Day 1" becomes obsolete by "Day 10" as the neural representation evolves.
2. **Recalibration Burden:** To counteract drift, traditional iBCI systems often require daily retraining or recalibration. This process is time-consuming and mentally exhausting for users, significantly reducing the clinical viability and "plug-and-play" utility of the device.
3. **Generalization Challenges:** Drift complicates the development of models that can generalize across long time scales. It necessitates advanced signal processing techniques, such as unsupervised domain adaptation [13] and manifold alignment [2], to track the stable latent dynamics from the shifting neural population.

The gap between the inconsistency in neural representations and the stability in behaviors requires more investigation.

1.2 Neural Population and Null-space

Recently, one important explanation on how behavior emerges from unstable neural representations is that instead of being encoded by the individual cells, the behaviors are encoded by the collective dynamics of the neural population.

1.2.1 Neural Population Perspective

Instead of analyzing neurons in isolation (e.g., "Neuron A encodes leftward movement"), the instantaneous activity of N recorded neurons can be viewed as a high-dimensional state vector, $\mathbf{x}(t) \in \mathbb{R}^N$. In this view, the behavior is not encoded by any single neuron firing rate, but by the pattern of co-activation across the entire population [1].

1.2.2 Neural Null-space

The mapping from the high-dimensional neural population in $\mathbb{R}^N$ space to low-dimensional behavior space, which has only very small number of degree of freedom ($\mathbb{R}^K$, where $K \ll N$), induced a massive reduction in dimensionality and information, which implies a high neural redundancy.

Mathematically, if we model a simple additive relationship between two neurons y_1 , y_2 , and their behavior readout x :

$$x = y_1 + y_2$$

the neural space can be decomposed into two orthogonal spaces, as illustrated in Figure 3. When the neural population activity of y_1 and y_2 moves along the axis labeled as null, the low-dimensional behavior readout x doesn't change, and only when the neural population activity's change has components on the coding axis, the behavior will change. Therefore, the neural population change can be divided into a Coding and a Null-space [18].

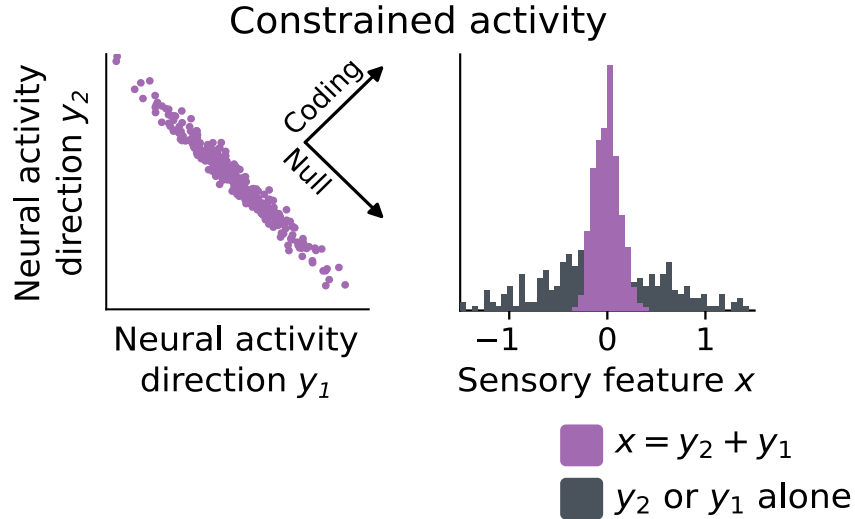


Figure 3: A simple example of neural population activity and low-dimensional readout.

1.2.3 The Null-Hypothesis for Drift

The decomposition offered a powerful explanation for the paradox of representational drift. The Null-space Hypothesis suggests that the brain allows representational drift to occur, provided that this drift is largely confined to the Output-Null space [6]. Even though individual neurons could contribute to both coding and null-space, as their tuning curves drift, the plastic mechanism in the network acts to constrain this drift by adjusting other neurons around this drift to maintain the population activity drift in the null-space.

However, as the Null-Hypothesis is getting more widely accepted, the physiological mechanism of this null-space population regulation remains unclear. In this paper, I’m going to discuss astrocyte cells’ potential role in regulating the neural population drift in the null-space to close this loop.

2 Astrocyte Regulation Confines Representational Drift to Null-space

2.1 Astrocyte Properties

Astrocytes are the most abundant glial cells in the mammalian brain. Astrocytes are named for their star-like shape, featuring thousands of fine processes. Astrocytes exhibit "tiling" – they arrange themselves in non-overlapping domains, and each astrocyte has exclusive control over a specific, immense territory, covering up to 2 million synapses. This tiling property ensures that the astrocytic regulation is localized and acts as an ideal "integrator" to detect large-scale network drift that individual neurons cannot perceive.

2.2 Astrocyte Functions

Historically, astrocytes were viewed merely as passive support cells that provided structural scaffolding and metabolic support to neurons as the "neural glue". However, modern neuroscience has shifted to the "Tripartite Synapse" model, which establishes that astrocytes are active participants in information processing [15]. There are key astrocyte functions relevant to neural dynamic regulations:

- **Neurotransmitter Clearance:** Astrocytes could rapidly remove glutamate from the synaptic cleft to prevent neurons from overactivation caused by the accumulation of glutamate in synapses, and maintain temporal precision. Knocking off the glutamate transporters on astrocytes will lead to the overactivation of the nearby neurons and failure in memory formation and learning [4, 7].
- **Gliotransmission** Astrocytes can release chemical signals, such as ATP, D-serine, that directly modulate synaptic plasticity, in response to neuronal activities. d-Serine released by astrocytes could increase the neuron excitability and promote the learning process [10], while the astrocytic ATP/Adenosine enables heterosynaptic suppression to stabilize networks [17].
- **Potassium Spatial Buffering:** Studies have shown that astrocytes can act as a spatial buffer to transport potassium ions, which participate in the formation of the action potential and are important in the regulation of synapse weights, from the high-concentration area to the low-concentration area, thus balance the global electrical potential and excitability of the network [14].

From these functions, it is obvious that astrocytes are able to participate in neural information processing and regulate neural population activities. In the context of representational drift, it’s easy to hypothesize that astrocytes act as a biological feedback control loop that maintains the excitation-inhibition balance of the network.

2.3 Astrocytic Regulation on Representational Drift

From the properties and functions of astrocytes we mentioned above, there might be two constraints on the neural population drift applied to neurons by astrocytes:

1. **Population Drift Direction Constraint:** Astrocytes constrain the neural population drift direction by activating the Hebbian learning of the entire local neuron ensemble to readjust the tuning and weights, so the population readout of this whole neural ensemble stays constant. The wide spatial range of the astrocytes may also allow the astrocytes to read the output of the entire neural ensemble directly; therefore, astrocytes could make sure the output of the neural ensemble stays unchanged by adjusting the local neurons and monitoring the direct readouts. This role is similar to hub neurons that regulate the neuron population dynamics [8], but the longer effect time and time constant make astrocytes better regulators for this process [7]. To physiologically achieve these functions, one possible

pathway is that astrocytes could react to the neuron activities by changing the free calcium Ca^{2+} concentration in the corresponding astrocyte process, which induces calcium waves that could spread to the nearby astrocyte processes, causing them to release gliotransmitters and finally promoting the learning process in the nearby neurons [16].

2. **Single Neuron Anchoring:** Astrocytes may act as anchors to keep a single neuron from drifting too far from its original representation. While individual neurons may drift, such as changing their firing rate or tuning curves, astrocytes that are directly monitoring their output synapses can sense local hyperactivity through their widespread processes and release gliotransmitters to suppress synaptic gain or upregulate glutamate transporters to dampen excitation. The longer effective scale period of astrocytes [7] may store the previous neuron firing pattern and work as a feedback regulator to anchor the single neuron firing pattern from drifting too much.

These two processes, facilitated by astrocytes, work together to keep the animal’s behavior consistent with the neural representational drifts. Part of the process is illustrated in Figure 4.

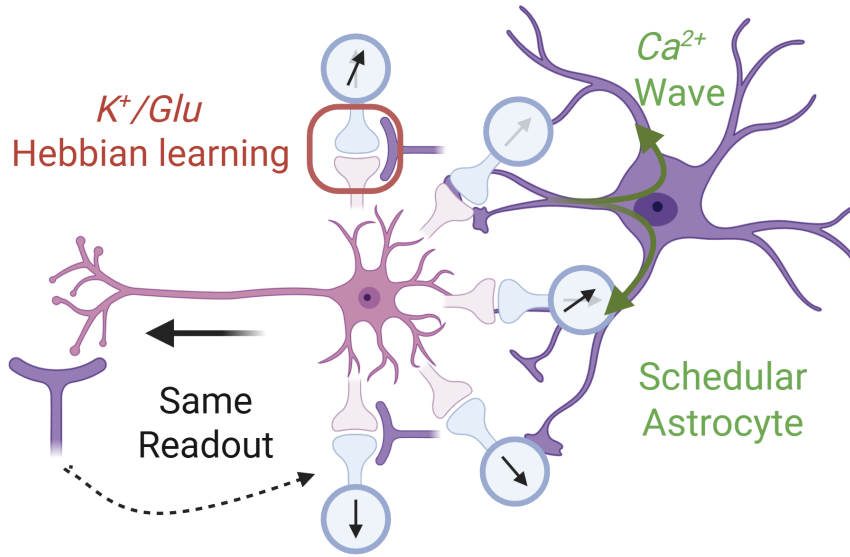


Figure 4: Astrocytes regulate the local neural ensemble in response to the single-neuron representational drift. Astrocytes regulate the local neural ensemble in response to the single-neuron representational drift. When a neuron drifts out of the recruited encoding subset, Scheduler Astrocyte could facilitate the Hebbian learning process by regulating the excitability of the nearby neuron to compensate, maintaining the same downstream readout.

3 Astrocyte Regulates Representational Drifts through Neuron Ensembles

3.1 The Excitability-Drift Hypothesis

Beyond their potential role in constraining population dynamics within the Output-Null space, astrocytes actively modulate the intrinsic excitability of individual neurons—a factor increasingly recognized as a primary driver of representational drift [3].

Current hypotheses suggest that representational drift is not a purely stochastic process but is driven by slow, continuous fluctuations in single-neuron excitability. Neurons operate with internal periods of volatility; their responsiveness to synaptic inputs is not static but follows a slow-timescale oscillation. As illustrated in Figure 5, this creates a dynamic "recruitment bias": at any given time, only the subset of neurons currently exhibiting high intrinsic excitability is recruited into the functional encoding ensemble. As the excitability

profile of the population shifts over days, the active subset changes, resulting in the observed drift of neural representations.

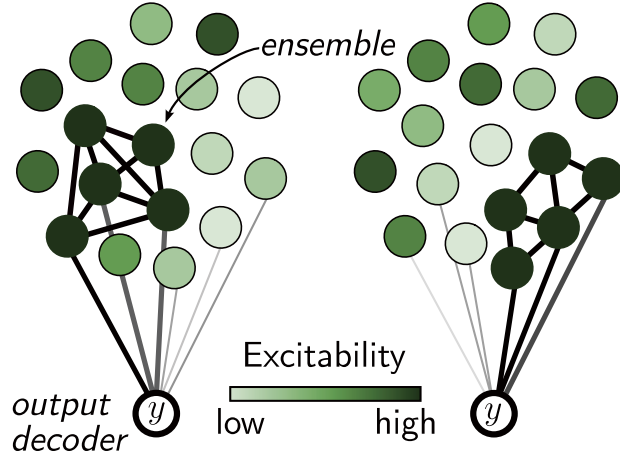


Figure 5: The same neuron ensemble at different times, only neurons with high excitability are recruited for behavior encoding.

3.2 Astrocyte as Excitability Scheduler

I propose that this regulatory function is mediated by a specialized operational mode of glia: the **”Scheduler Astrocyte”**. Just as recent network analyses have identified **”hub neurons”**, which are highly interconnected neurons that play a disproportionate role in orchestrating local circuit dynamics and information flow [8], we hypothesize a glial counterpart. A Scheduler Astrocyte physically enwraps and functionally binds a specific neuronal ensemble responsible for a specific behavior within its non-overlapping territorial domain.

In this view, the Scheduler Astrocyte does not merely support individual cells but acts as a local governor for the entire ensemble. By controlling the uptake of glutamate via GLT1 transporters, it effectively tunes the **”gain”** or plasticity of its specific local circuit. This regulatory mechanism can be analogized to a learning rate scheduler in machine learning optimization algorithms, such as Stochastic Gradient Descent (SGD).

A recent study in Sur’s lab provided compelling evidence for this computational role. [4] demonstrated that deactivating the astrocyte glutamate transporter (GLT1)—essentially breaking the **”Scheduler”**—leads to unregulated, persistently high neuronal excitability. As illustrated in Figure 6, the GLT1-deficient mice exhibited significantly lower response times during intermediate training (Sessions 6–10), indicating a faster learning speed compared to healthy controls. However, this rapid acquisition came at a cost: the inter-trial motor trajectory similarity and smoothness were significantly lower.

This phenomenon parallels the behavior of a deep learning model trained with a consistently high learning rate:

- **High Excitability (Scheduler-Deficient) $\approx$ High Learning Rate:** Without the Scheduler Astrocyte’s damping, the system updates weights rapidly (fast learning) but **”overshoots”** or oscillates around the global minimum, resulting in noisy, unstable trajectories.
- **Regulated Excitability (Healthy Scheduler) $\approx$ Scheduled Decay:** In healthy subjects, the Scheduler Astrocyte dynamically dampens excitability as the task is learned. This acts like a scheduler that anneals the learning rate as the loss decreases, ensuring that the optimization process slows down to allow the neural manifold to converge to a smooth, stable, and robust solution.

Therefore, by modulating the recruitment of neurons through excitability changes, Scheduler Astrocytes essentially balance the trade-off between exploration (neural plasticity) and stability during the learning process.

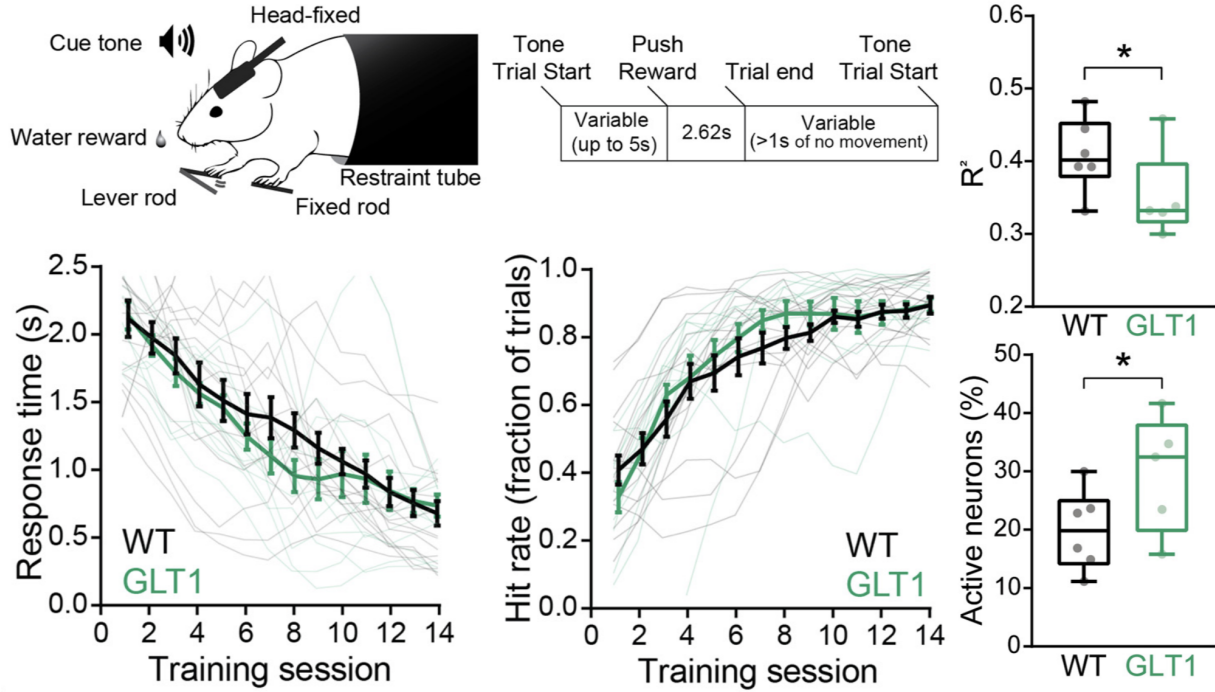


Figure 6: Astrocytes act as "learning rate schedulers" during motor learning. Deactivating the glutamate transporters in astrocytes leads to a faster but more unstable learning process. This plot is from an astrocyte study from Sur's lab at MIT [4].

3.3 Mechanism for Drift Control

Crucially, combining with the **Excitability-Drift Hypothesis**, this "Scheduler" capability can be extended beyond initial learning to the continuous maintenance of the neural code. We postulate that the Scheduler Astrocyte imposes **Homeostatic Gain Control** on the population level to counteract stochastic drift:

- **Hub-Mediated Integration:** Similar to how hub neurons integrate inputs to synchronize network activity [8], the Scheduler Astrocyte integrates metabolic and synaptic signals from its entire ensemble. It buffers extracellular K^+ and clears glutamate to clamp the physiological range in which the *collective* ensemble can fluctuate, preventing intrinsic oscillations of single neurons from pushing the population state out of the stable Null-space.
- **Compensatory Normalization:** The Scheduler Astrocyte functions as a biological layer normalization mechanism [15]. Because it monitors the aggregated activity of the local ensemble, it can enforce a conservation of total excitability. When a subset of neurons inevitably "drifts out" (lowered excitability), the Scheduler Astrocyte senses the resulting drop in local network summation and compensatorily increases the excitability of neighboring neurons within the same ensemble. This hub-like regulation ensures that the low-dimensional behavioral readout remains consistent over time, even as the constituent neurons participating in the ensemble change.

4 Potential Experimental Validation

There are two key points that could be tested to solidify this theory:

4.1 Deactivating Astrocytes Leads to Catastrophic Forget

To empirically validate the hypothesis that astrocytes actively regulate representational drift and confine it within the output-null space, I propose extending the experimental paradigm established by the Sur lab [4]. If the hypothesis is correct, disrupting astrocytic regulation should uncouple neural drift from behavioral stability. Consequently, we predict that deactivating astrocytic function will cause the neural population to drift out of the null-space, resulting in rapid behavioral regression (catastrophic forgetting).

The proposed experimental design is as follows:

1. **Acquisition Phase:** Train two cohorts of mice (Control vs. Experimental) on an operant conditioning task, such as a lever-press water reward task. Training continues until both groups achieve stable performance, characterized by a plateau in hit rate, reaction time, and kinematic stereotyping.
2. **Manipulation and Retention Interval:** Following mastery, cease all training to prevent further continuous learning. In the **Experimental Group**, inhibit astrocytic function (specifically targeting GLT1 glutamate transporters or using chemogenetic silencing). The **Control Group** receives a sham manipulation. Both groups are then housed normally for a defined retention period to allow natural representational drift to occur.
3. **Cross-Sectional Assessment:** To assess memory retention without the confounding factor of "re-learning" during testing, a cross-sectional approach is employed. At periodic intervals (e.g., Day 3, 7, 14), a distinct subset of mice from each group is subjected to a recall test. Following this single test session, these subjects are excluded from subsequent time-point analyses.

Expected Outcome: I hypothesize that the Experimental Group (GLT1-deficient) will exhibit a significantly faster rate of performance decay compared to the Control Group. In the absence of astrocytic regulation, we postulate that stochastic fluctuations in neuronal excitability will drive the population activity out of the output-null space. This unconstrained drift will degrade the neural representation of the task, manifesting as a failure to retrieve the learned behavior despite the absence of external interference.

4.2 Deactivating Scheduler Astrocytes Impairs Specific Behavioral Encodings

Under the hypothesis that a "Scheduler Astrocyte" and its physically coupled neuronal ensemble form a coherent functional unit responsible for a distinct behavioral feature, such as encoding a visual orientation or a specific movement vector, we predict that the selective deactivation of the Scheduler will result in the collapse of that specific behavioral output.

Previous research supports the existence of such functional specificity; astrocytes in the visual cortex have been shown to exhibit sharp, stimulus-dependent tuning curves akin to neuronal representations [19]. We posit that these highly tuned, slow-integrating glial cells are the biological instantiation of the "Scheduler Astrocytes" proposed in our model. Assuming an analogous functional architecture exists in the motor cortex, these astrocytes would serve as hubs for encoding specific motor primitives. Consequently, their targeted inactivation is expected to impair the specific movement functions they regulate, decoupling the neural ensemble's stability and degrading the behavioral readout.

5 Potential Applications in iBCIs: Towards a Hybrid Architecture

Recent breakthroughs have challenged the neuron-centric view of cortical coding. Studies demonstrate that astrocytes exhibit tuned responses analogous to neuronal cosine tuning, with evidence showing even sharper directional selectivity in the visual cortex (V1) compared to neurons [19]. With the advent of advanced optical interfaces for calcium imaging, utilizing astrocytic signals for next-generation iBCIs becomes a theoretical possibility. However, relying solely on astrocytes presents significant engineering hurdles:

- **High-Dimensional Signal Multiplexing:** Astrocytes exhibit "mixed selectivity," encoding not only instantaneous kinematic variables (e.g., preferred direction) but also integrating information regarding behavioral history, reward prediction errors, and metabolic demand [16]. Although information redundancy reveals a more accurate neural state, algorithms must separate the simple true motor intent

from history effects, making this redundancy double-edged. The encoding of trial success or error information could effectively serve as an intrinsic "ground truth" signal for reinforcement learning-based BCI adaptation.

- **Prohibitive Temporal Latency:** The most critical limitation is the timescale mismatch. Astrocytic calcium transients operate on a timescale of seconds, with response lags up to 4–5 seconds relative to stimulus onset [16]. This latency renders an astrocyte-only system incapable of closing the loop for real-time, high-bandwidth motor control (e.g., cursor movement), which requires millisecond-level precision.

To bridge this gap, I propose a **Neuron-Astrocyte Hybrid BCI (NAH-BCI)** architecture. Rather than treating astrocytes or neurons as a direct source of kinematic control signals, this system utilizes astrocytes as a "**Contextual State Estimator.**"

In this framework:

1. **The Fast Stream (Neurons):** A traditional electrode array decodes high-frequency spiking activity for real-time motor execution.
2. **The Slow Stream (Astrocytes):** Simultaneous optical or chemical recording of astrocytes tracks the slowly fluctuating "neural state variables," such as alertness, fatigue, or plasticity levels.

Mechanism of Integration: Astrocytes function as a biological "meta-learner." Since they govern the recruitment of neural ensembles (as discussed in the Excitability-Drift Hypothesis), their activity pattern acts as a **Prior information**. By decoding the current astrocytic state, the BCI can predict *which* subset of neurons is currently highly excitable and likely to be recruited. This information allows the decoder to dynamically re-weight the neural inputs, ignoring the "drifted-out" neurons and prioritizing the "recruited" ones—thereby achieving long-term robustness without frequent recalibration.

6 Conclusion

In conclusion, the paradox of stable behavior emerging from drifting neural representations represents a fundamental challenge for both neuroscience and neuroengineering. This report proposes that this stability is not an intrinsic property of neuronal circuits alone, but a feature of **neuro-glial interaction**. Astrocytes act as the critical "manifold stabilizer" of the cortex by dynamically modulating intrinsic neuronal excitability and integrating local population activity. These mechanisms work in concert to confine stochastic neural drift within the null-space, ensuring that functional readouts remain robust despite the continuous change of individual neurons. The hypothesis could be validated through the proposed GLT1-manipulation experiments to provide the causal link between glial regulation and manifold stability.

Furthermore, translating these biological insights into engineering, through the **Neuron-Astrocyte Hybrid BCI** architecture, offers a promising path toward overcoming the "decoder degradation" challenge. Ultimately, shifting from a neuron-centric view to a holistic neuro-glial perspective is essential for decoding the brain's capacity for long-term resilience, building the next generation of robust brain-computer interfaces.

7 Six Questions

1. **What adaptive goal would this non-neural system serve by controlling behavior?**

Astrocytes, especially the **Scheduler Astrocytes** I came up with in this paper, regulate the neural population excitability and confine the subsequent dynamic activities in the null-space, so animals' behaviors are robust to the representational drift. Therefore, even though there is representational drift happening at the single-neuron level to facilitate continuous learning, the low-dimensional readout remains the same.

2. **What kind of dynamics do the non-neural systems display?**

The changes in neurotransmitter concentrations, such as glutamate, activate the astrocyte processes, facilitating the Ca^{2+} wave. The Ca^{2+} wave activates the nearby processes, causing them to release gliotransmitters and ions, facilitating Hebbian learning to reweight the synapses.

3. **What signals does it transmit to neural systems to create a behavioral impact?**

This system includes neurotransmitters, free ions, calcium waves, and gliotransmitters. The most important one is the calcium waves, which happen within the astrocyte cells to connect the local processes and broadcast the local neural information.

4. **Which neural system(s) does it impact?**

Both Representational drifts and astrocytes are widely spread in the brain. Therefore, the whole central system is impacted.

5. **How does it impact them (e.g., rate or synchrony changes)?**

The astrocytes release ions and gliotransmitters to regulate the excitability, which facilitates the Hebbian learning and determines if the neuron is recruited in the encoding process. This process happens on a time scale of hours or even days.

6. **What crucial tests would you conduct to test whether this system creates the proposed behavioral impact?**

As described in the paper, two important tests are first to deactivate the astrocytes, to see if the animal suffers from a catastrophic forget, and second is to deactivate an astrocytic ensemble that is selectively activated for a specific behavior, to see if selectively deactivating this ensemble would damage this behavior.

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